

Dataset Comparison: Grid & Household Electricity

This document outlines the straightforward differences between the 2024 StoreNet dataset and the newer 2026 dataset, detailing which is best suited for your initial ML demand-prediction prototype.

2024 StoreNet Dataset

Mainly focused on normal houses and their core electricity systems. It provides very detailed, minute-by-minute data during 2020. Highly suitable for predicting basic electricity needs ("How much electricity will this house need in the next hour?").

-  **House consumption:** overall electricity usage
-  **Solar/PV:** electricity produced by solar panels
-  **Battery:** charging, discharging, and levels
-  **Grid:** electricity taken from or sent to the grid
-  **Weather:** temperature, radiation, rain, wind

2026 Dataset

Also based on houses in Ireland, but features more advanced electrical equipment and cost factors. This dataset expands on the basic model by introducing heavy domestic loads and variable pricing models.

-  **Electric vehicle (EV):** electricity used for car charging
-  **Heat pump:** electrical system for heating/cooling
-  **Electricity tariff:** time-of-use pricing (e.g., cheap afternoons/nights, expensive evenings)

Feature Comparison Matrix

Feature	2024 StoreNet	2026 Dataset
Number of Houses	20	4
Data Interval	1 minute	30 minutes
House consumption	✓	✓
Solar panels	✓	✓
Battery	✓	✓
Grid connection	✓	✓
Weather data	✓	✓
Electric car charging 	✗	✓
Heat pump 	✗	✓
Electricity price (Tariffs) 	✗	✓

*In short: 2024 = basic version (houses + solar + battery + grid + weather).
2026 = advanced version (similar houses + electric cars + heat pumps + electricity prices).*

Project Recommendation: Start with 2024

For your initial project phase, it is highly recommended to start with the **2024 StoreNet dataset**.

Why? Your primary goal is to answer: **"Can we use real Irish data to train an ML model that predicts electricity demand?"** The 2024 dataset contains everything necessary to prove this concept, with highly